Portfolio 5 - Meta analysis of voice features in schizophrenia

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Experimental Methods 3

Due 11th of December 2020

Link to code: https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_5

Introduction

(Anders) This meta-analysis-paper seeks to compare different studies' results on predicting schizophrenia by the vocal feature of pitch.

Methods

(Sigrid) The full process of the literature review can be found in the PRISMA-chart in appendix A. For the quantitative synthesis, 48 articles were included. Some articles included several studies, so in total there were 50 studies. Out of these 50 studies, we conducted meta-analysis on 6 of the studies. These 6 studies all included quantitative measures for mean and standard deviation of pitch for both groups; healthy controls (HC) and schizophrenics (SZ). Many other variables than pitch have been proposed to diagnose schizophrenia. This meta-study focuses only on pitch as a predicting variable for schizophrenia.The 6 studies used for the meta analysis only represented (6/50 = 0.12) 12% of the overall studies reviewed.

The meta-analysis was conducted using the metafor package (Viechtbauer, 2010) in R (R Core Studio, 2019).

Variables

(Gustav) The mean and standard deviation of pitch were assessed for both the control and schizophrenic group across all studies. A standardized effect size and the associated variance was calculated by use of the escalc() function from the metafor package (Viechtbauer, 2010).

Analysis and models

(Anders) The analysis was conducted on both the original dataset consisting of six studies, and on a dataset including the studies from assignment 3. These datasets will from here on be referred to as *the original dataset* and *the full dataset*. The data from assignment 3 consists of 4 studies, which we have pooled into one study.

The models were made using the rma() function from the metafor package (Viechtbauer, 2010). The rma() function creates a version of a general linear mixed effects model, which is suitable for meta analyses. The model only consisting of random effects was written using the following syntax:

$$m_{random} <- rma(yi, vi, data, slab = StudyID)$$

An additional model with fixed effects was made. The model was built using the following syntax:

$$m_{fixed} <- rma(yi, vi, mods = cbind(TYPE\ OF\ TASK), data, slab = StudyID)$$

Forest-plots, bajaut-plots and funnel-plots were made using the functions forest(), bajaut() and funnel(), respectively, from the metafor-package (Viechtbauer, 2010). The functions regtest() and ranktest() were used to calculate objective measures for funnel-plot asymmetry, also from the metafor-package.

Results

Results from the fixed effect model

(Sigrid) Our fixed effect model on the original dataset showed no significant effect of the moderator ‘type of task’ with QM(df = 1) = 0.5154, p-val = 0.4728. This was also the case for the dataset where study 3 had been added with QM(df = 1) = 0.5952, p-val = 0.4404. The analysis thereby indicates that the type of tasks condition does not account for any of the heterogeneity in the data.

Results from the original dataset

(Gustav) All the following results are from the random effects model. The meta-analysis on the original dataset indicates that the studies were not heterogeneous, Q(df = 5) = 9.8472, p-val = 0.0797. The Tau^2 statistic is also indicative of no heterogeneity, tau^2 = 0.07 (SE = 0.0908). The I^2 statistic was 50.29% which represents a moderate variance between the studies. The summary effect size was non-significant at -0.16 CI[-0.46,0.14], SE = 0.15, $p > 0.05$ ($p = 0.29$). The forest plot can be found in figure 1.

Egger's regression test indicated funnel-plot asymmetry, z = -1.58, p = 0.11. The funnel plot can be found in figure 2. Kendall's rank correlation test also indicated funnel-plot asymmetry, $\tau = -0.20$, p = 0.72.

Since the studies were found to be relatively homogenous and the full dataset only contains 6 studies, we will not remove the most influential studies found in the upper right quadrant of the Baujat-plot (Figure 2).

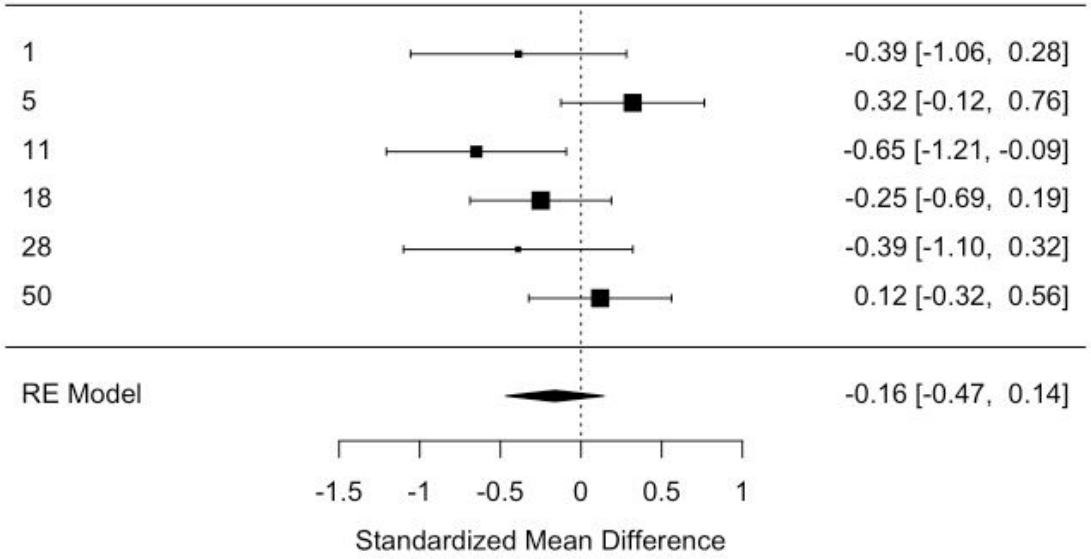

Figure 1 Forest plot for the original dataset

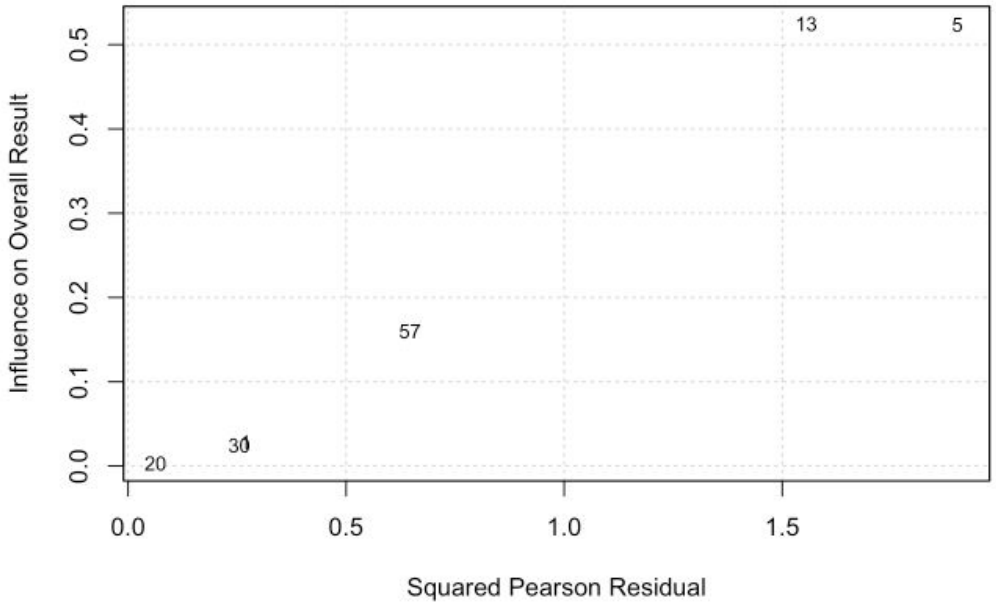

Figure 2: Baujaut plot for the original dataset

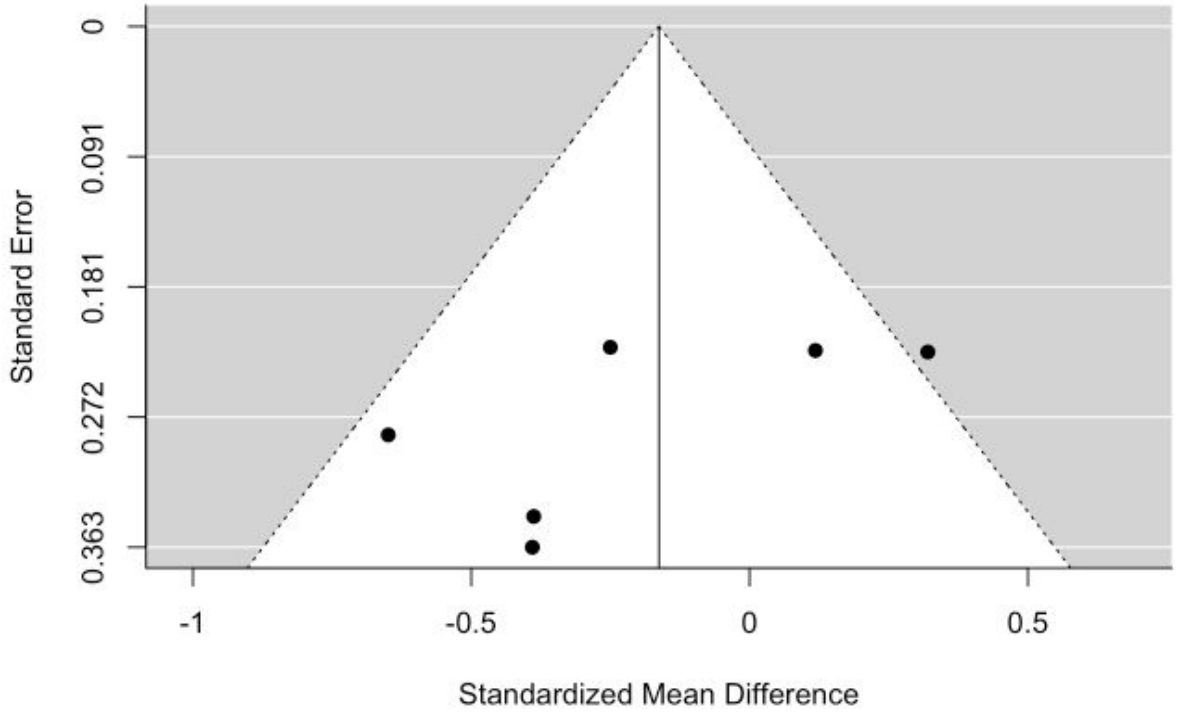

Figure 3: Funnel plot of the original dataset

Results from full dataset

(Anders) All the following results are from the random effects model made from the full dataset.

The output of the random effects model indicates that the studies were not heterogeneous, Q (df = 6) = 9.96, p = 0.13. The amount of variance which could be attributed to heterogeneity between studies rather than within studies was found to be low to moderate, I^2 = 41.42 %. The total amount of heterogeneity was calculated using tau^2, tau^2 = 0.04, SE = 0.05. The summary effects size was non-significant at -0.16, CI[-0.38, 0.07], SE = 0.11, p > 0.05 (p = 0.17).

Egger's regression test showed funnel-plot asymmetry, z = -0.81, p > 0.05 (p = 0.42).

Since the studies were found to be relatively homogenous and the full dataset only contains 7 studies, we will not remove the most influential studies found in the upper right quadrant of the Baujat-plot (Figure 5).

Kendall's tau rank test also showed funnel-plot asymmetry, τ = -0.24, p > 0.05 (p = 0.56).

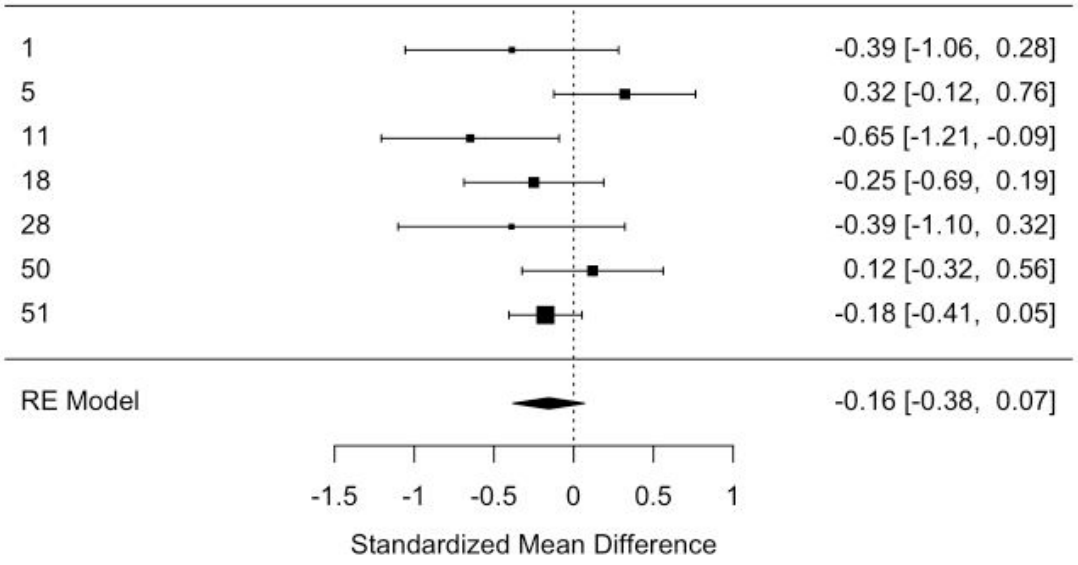

Figure 4: Forest plot for the full dataset. Study 51 indicates the newly added study

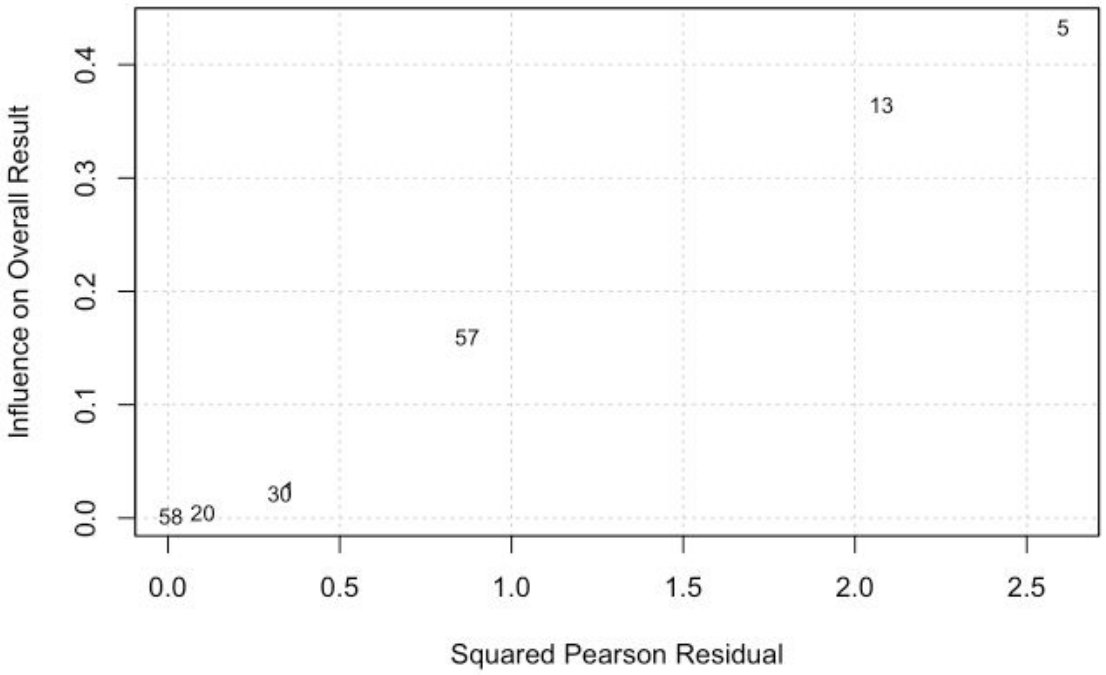

Figure 5: Baujat plot for the full dataset.

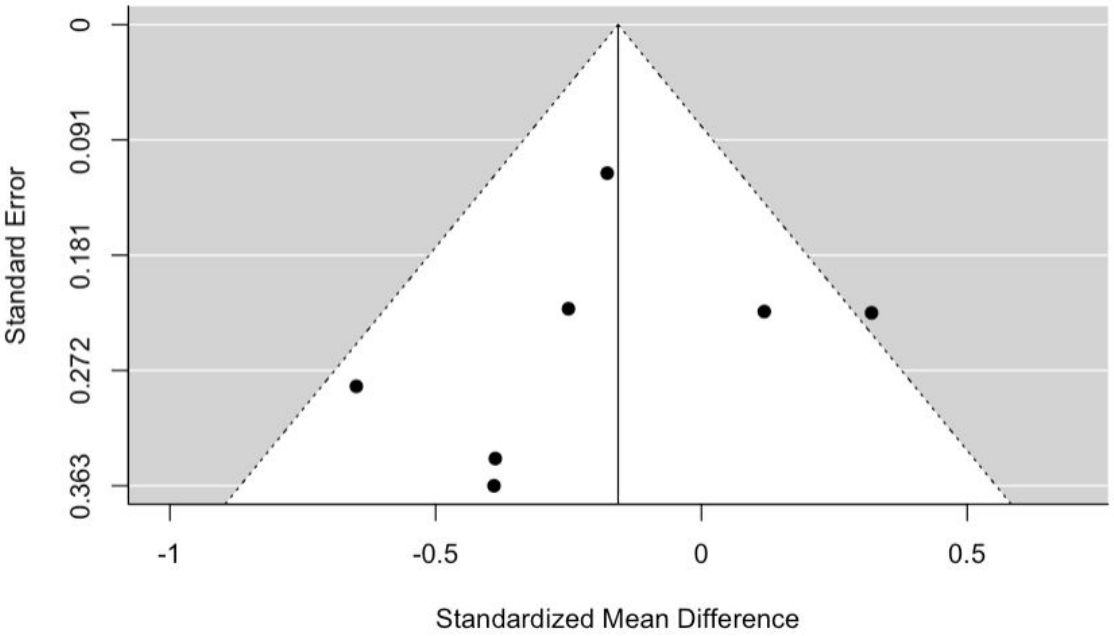

Figure 6: Funnel plot of the full dataset.

Discussion

(Sigrid) The results from the original dataset suggested that the studies were not heterogenous. Thus, the studies were eligible to conduct a meta-analysis. The studies summary effect size was calculated and found to be non-significant. The tests investigating funnel-plot asymmetry found that there might be a publication bias, since the funnel-plots were significantly asymmetrical.

The results from the full dataset showed that the studies were not heterogenous. The summary effect size was, however, not significant, thereby indicating that there is no common effect size of the studies. The tests investigating funnel-plot asymmetry found that there might be a publication bias, since the funnel-plots were significantly asymmetrical.

Although, the numerical tests for publication bias for both the original and the full dataset showed funnel-plot asymmetry, we question the validity of these tests. The funnel plots only include respectively 6 and 7 studies which makes it difficult to determine anything about the distribution of the studies in the funnel plot. Likewise, it is not possible to determine whether a distribution is gaussian with only 6 or 7 data points. Therefore, we conclude that it is not possible to determine whether there is a publication bias in the data because of the low number of studies included in the analysis.

(Gustav) Adding study 3 to the original set (the full dataset) improved the results of the analysis. Tau^2 dropped from 0.07 to 0.04 indicating that the total amount of heterogeneity in the dataset decreased after adding study 3. Also, the I^2 decreased from 50.29 % to 41.42 %, indicating that less of the variance in the data was caused by heterogeneity, but rather by with-in study variance. Adding study 3 thereby resulted in less heterogeneity in the meta-analysis.

The summary effect size was still insignificant after adding study 3. However, the p-value did improve (from $p = 0.29$ to $p = 0.17$), indicating that the summary effect size was closer to being marginally significant. Thus, adding study 3 improved the non-heterogeneity and the p-value of the summary effect size. We propose that adding more studies to the meta-analysis might result in a significant summary effect size and a more accurate estimate of publication bias. It seems that our meta-analysis lacks power because of the low sample size, but there might still be a true underlying effect.

References

R Core Team (2019). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL https://www.R-project.org/.

Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. Journal of Statistical Software, 36(3), 1-48. URL: https://www.jstatsoft.org/v36/i03/

Appendix A

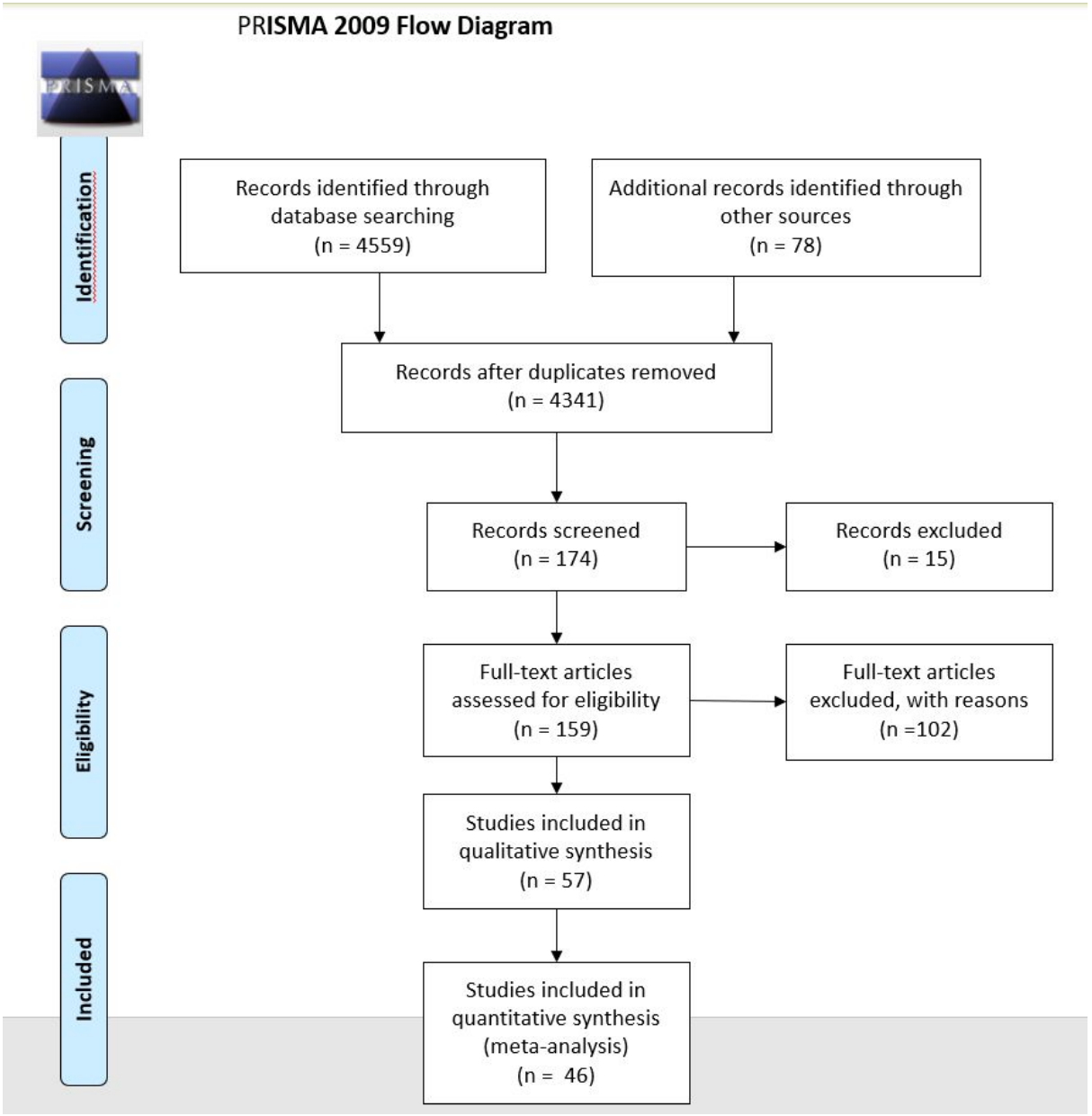

Figure A: PRISMA-chart depicting the literature-search process.